

AcePharm — Web Platform: Project Tracker

Production Delivery & Milestone Execution Tracker (Next.js + Prisma / Postgres + pgvector)

This document tracks the seven core development milestones and QA gates for the live **AcePharm** web revision platform (acepharm.co.uk). Follow and check off each stage sequentially to verify data model integrity, grounded AI citations, sub-300ms question interactions, and compliance with WCAG 2.2 AA standards.

Milestone 1: Foundations, Data Model & RBAC Setup

- **Repository & Environment Scaffolding:** Configure Next.js 14+ (App Router, Strict TS), Tailwind CSS, shadcn/ui tokens, Sentry error logging, and PostHog analytics (EU region).
- **Postgres & pgvector Schema:** Establish Prisma ORM schema covering User, Session, Question, Category, Attempt, Note, Bookmark, and embedding vectors. Configure UK/EU hosting.
- **Dual-Store Attempt Architecture:** Implement immutable first-attempt store (QuestionFirstAttempt) isolated from repeatable practice history (QuestionAttempt) with composite indices.
- **Native Authentication & Role Guards:** Implement Auth.js credentials and magic-link authentication with server-side role enforcement (Admin, Author, Reviewer, Learner) returning HTTP 403 upon denied access.
 - **Test Environment: Local Development & Staging Postgres DB (verify schema migrations, role gate unit tests, and Auth.js session cookies).**

Milestone 2: Content Pipeline, Editorial Workflow & Importer

- **Curriculum Hierarchy Manager:** Structure high-yield BNF therapeutic systems, GPhC exam weightings, high-risk drug flags, and calculation topic models.
- **Clinical Question Editor:** Build authoring UI enforcing stem, 5 distinct options, individual rationales per option, subtopic clinical notes, and 'AcePharm Clinical Editorial Team' attribution.
- **Question Review & Audit Log:** Implement formal status transitions (Draft → Under Review → Approved → Published) with full author/reviewer version history.
- **CSV / Spreadsheet Importer:** Deploy batch question importer with dynamic column mapping, syntax validation, duplicate detection, and pre-import error reporting. Ingest Seed Bank v0.1 (135 questions).
 - **Test Environment: Headless validation runner & Staging Admin Portal (verify seed data ingestion and mandatory option rationale validation).**

Milestone 3: Question Practice Engine (Core Loop)

- **Session Builder:** Configurable question sessions filtered by category, high-risk flags, difficulty, past performance (All, Unused, Incorrect, Flagged), and mode (Learn vs. Timed).
- **Strict Answering Rules:** Enforce pre-submission confidence capture (Low/Medium/High) without color signaling. On submit, reveal accessible SVG icons, textual status, and per-option rationales.
- **Clinical Feedback & Interaction Tools:** Subtopic revision notes, hide-options mode (per-user preference), flagging, personal notes, and instant report-a-question modals with admin routing.
- **High-Performance Execution:** Ensure sub-300ms perceived answer submission via optimistic UI reconciliation and background prefetching of adjacent question stems.
 - **Test Environment: Desktop & Mobile Browsers (Chrome, Safari, Firefox) verifying 20-question session execution, timer integrity, and dual-attempt logging.**

Milestone 4: Progress Analytics & Recommendation Engine

- **Five Uncollapsed Metrics:** Compute and display First-Attempt Accuracy, Practice Accuracy, Repeat Accuracy, Confidence Calibration, and Curriculum Coverage without collapsing into a single metric.
- **Visual Topic Map:** Implement 6-tier progression taxonomy: Not Started → First Pass → Needs Attention → Developing → Secure → Due for Review.
- **Smart Recommendation Engine:** Generate defensible next practice recommendations with visible clinical rationale based on confidence discrepancies and spaced review intervals.
- **Category Reset Isolation:** Verify that learner category resets purge active practice history while leaving pristine first-attempt benchmarks permanently intact.
 - **Test Environment: Staging Analytics Environment (test with empty accounts, mid-tier accounts, and high-volume 100+ attempt seed accounts).**

Milestone 5: Grounded AI Tutor Layer — 'Ace'

- **Semantic Knowledge Chunking:** Index explanation rationales, subtopic notes, and BNF reference summaries using pgvector embeddings.
- **Strict Grounding Architecture:** Prompt orchestration via Anthropic Claude (Messages API) generating answers strictly from retrieved AcePharm content with verifiable citations. Zero hallucination.
- **Deflection & Safety Guardrails:** Explicit graceful refusal for non-covered topics or real-patient diagnostic queries. Log cost, token usage, and chunk IDs per message.
- **Specialized Ace Modules:** Highlight-to-ask inline queries, calculation coach with step-by-step guidance, and spaced-repetition flashcard generator with fair-use rate limits.
 - **Test Environment: Automated 50+ Case Clinical Evaluation Suite (benchmark citation accuracy, refusal of out-of-domain prompts, and patient safety checks).**

Milestone 6: Public Marketing Site & Stripe Subscriptions

- **Public Marketing & SEO Architecture:** Build Homepage, Features, Question Bank, About, Editorial Standards, FAQ, Contact, and Legal Hub with SSR, metadata, and structured schema.
- **Editorial Content & Blog CMS:** Simple admin-managed blog engine with scheduled publication gating and strict compliance auditing (no unapproved competitor claims).
- **Stripe Billing Integration:** Deploy Stripe Checkout and Customer Portal supporting Explorer (30 free Qs), Monthly (£16/mo), and Annual (£120/yr) plans with webhook signature validation.
- **Frictionless Online Cancellation:** Self-service cancellation retaining account access until end of current billing period, permanently preserving learner history in locked state.
 - **Test Environment: Stripe CLI & Staging Webhook Listener (simulate complete checkout, invoice renewals, payment failures, and tier downgrades).**

Milestone 7: Hardening, WCAG 2.2 AA & Production Launch

- **Full Accessibility Audit:** Verify WCAG 2.2 AA across all screens: keyboard navigation, visible focus indicators, screen reader live regions, and colorblind contrast.
- **Security & Rate Limiting:** Argon2/bcrypt password hashing, CSRF tokens, strict CSP headers, and rate limiting on auth, question reporting, and AI endpoints.
- **Disaster Recovery Verification:** Conduct automated database backup and test complete staging database restoration from snapshot.
- **Submission Documentation:** Complete 11-point Milestone Verification Document, QA checklist, and record comprehensive Loom walkthrough video.
 - **Test Environment: Automated axe-core / Lighthouse audit (target 90+ across all metrics) & Production Staging Environment.**

Project Links & Distribution Status

- **GitHub Repository:** <https://github.com/takweentutors1/acepharm-web-app> (main branch)
- **Staging Preview URL:** <https://staging.acepharm.co.uk> (Vercel Preview Deployment)
- **Stripe Test Dashboard:** Configured in Test Mode (Explorer, Monthly £16, Annual £120)
- **Loom Walkthrough Video:** Record 5–8 min milestone walkthrough upon final delivery
- **PostHog & Sentry Monitoring:** Active (EU Region Data Isolation)